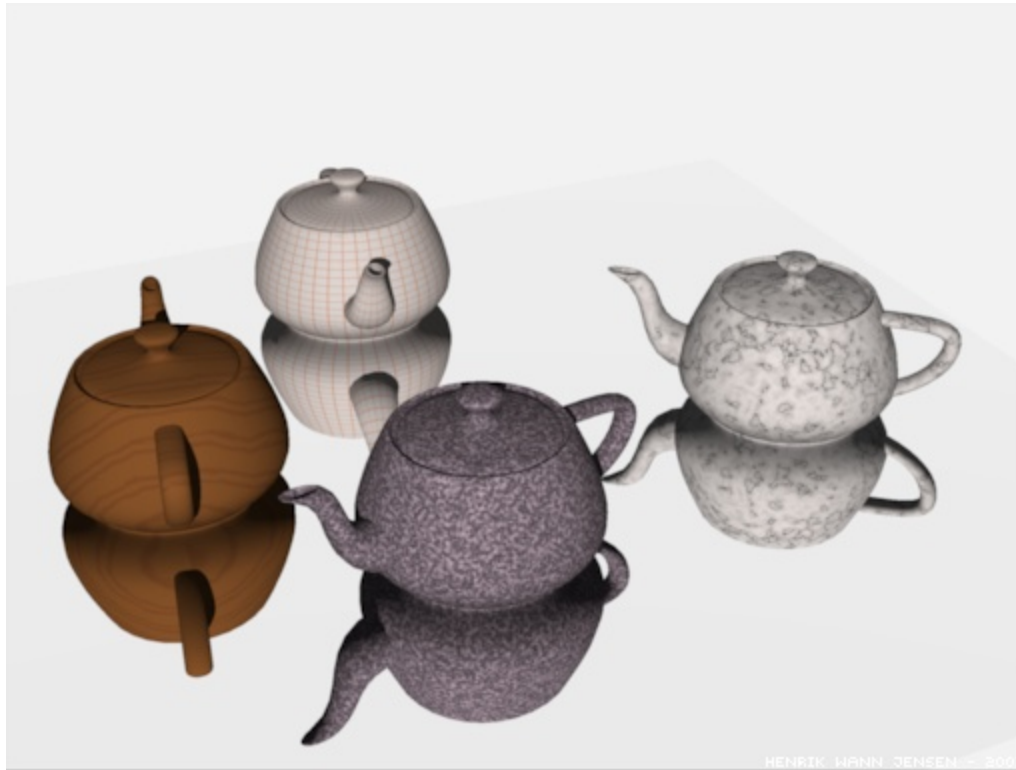


CSE168

Computer Graphics II, Rendering



Spring 2006
Matthias Zwicker

Last time

- Path tracing and the measurement equation
- Photon mapping introduction

Measurement equation

- If we have random variable $E[\tilde{L}_i] = L_i$ then we can compute any measurement

$$\begin{aligned} I &= \int \int W(\mathbf{x}, \omega) L_i(\mathbf{x}, \omega) d\omega d\mathbf{x} \\ &= E \left[\frac{1}{N} \sum_j W(\mathbf{x}_j, \omega_{i,j}) \tilde{L}_i(\mathbf{x}_j, \omega_{i,j}) \right] \end{aligned}$$

for any importance function W

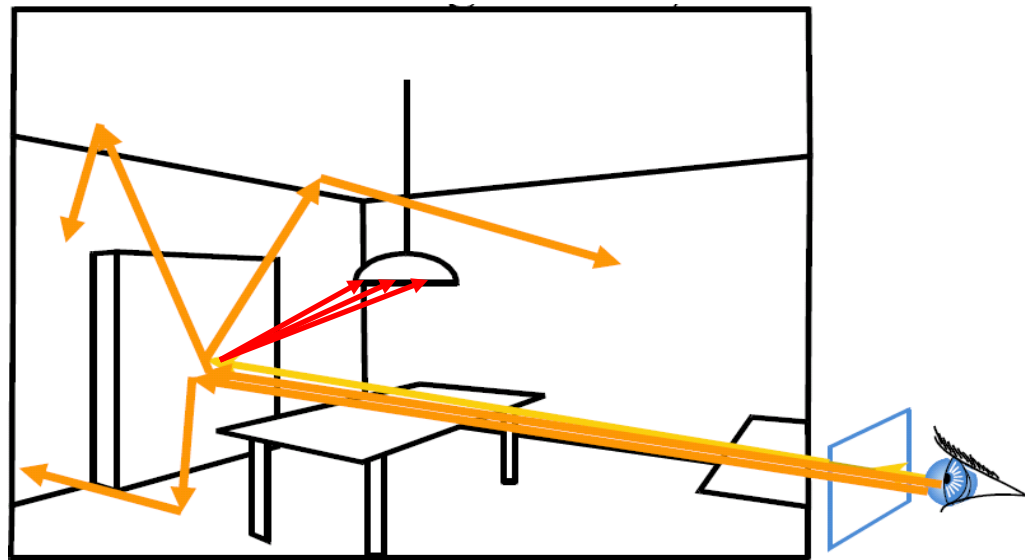
- Note: N is the number of paths

Path tracing

Bottom line: estimate pixel values as weighted average of random paths through pixel

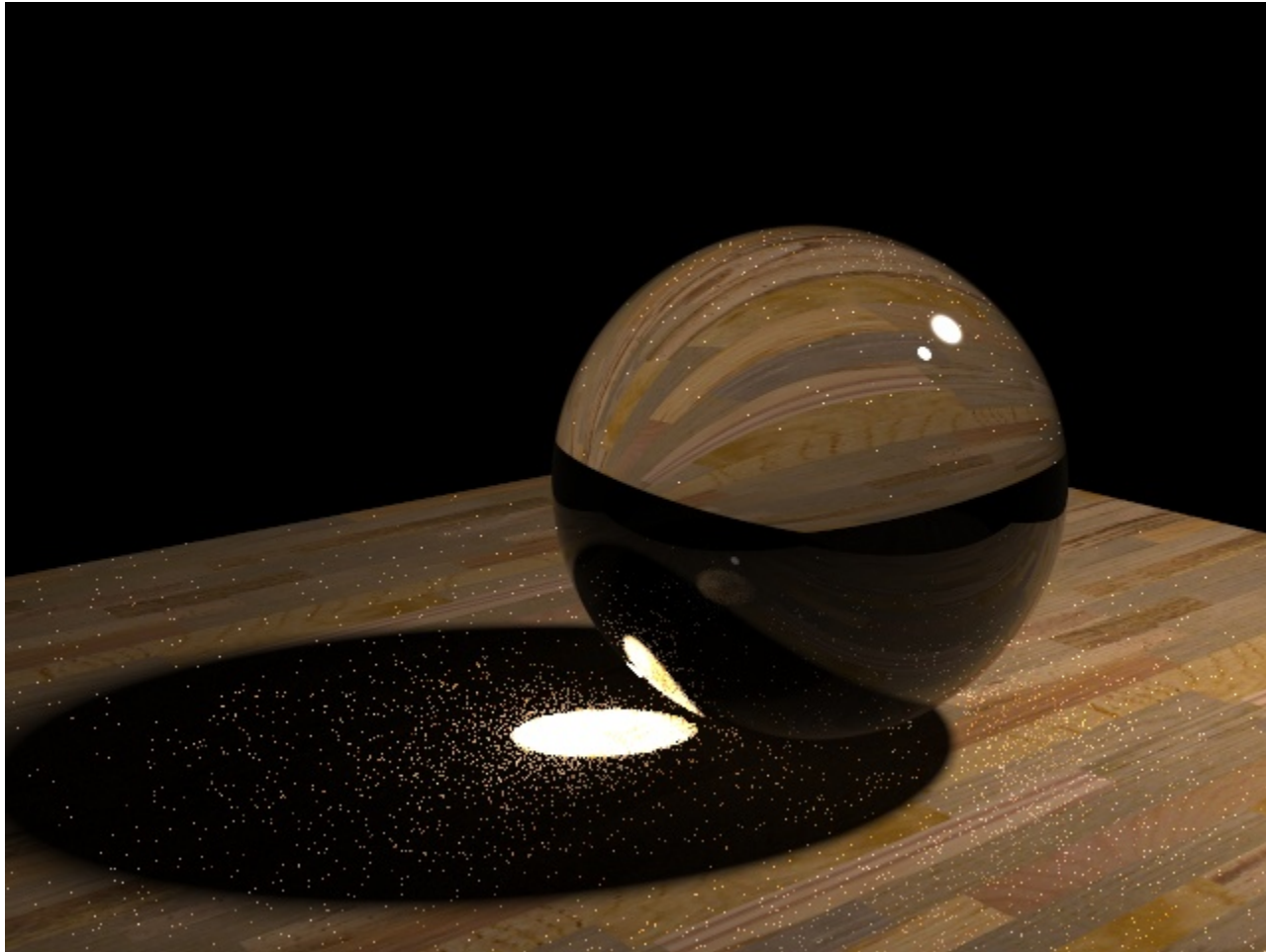
Path tracing algorithm

- Construct paths incrementally starting at the eye, dozens of rays for each pixel
- Shoot shadow rays at each path vertex
- Compute pixel as weighted average of paths



[Cutler, Durand]

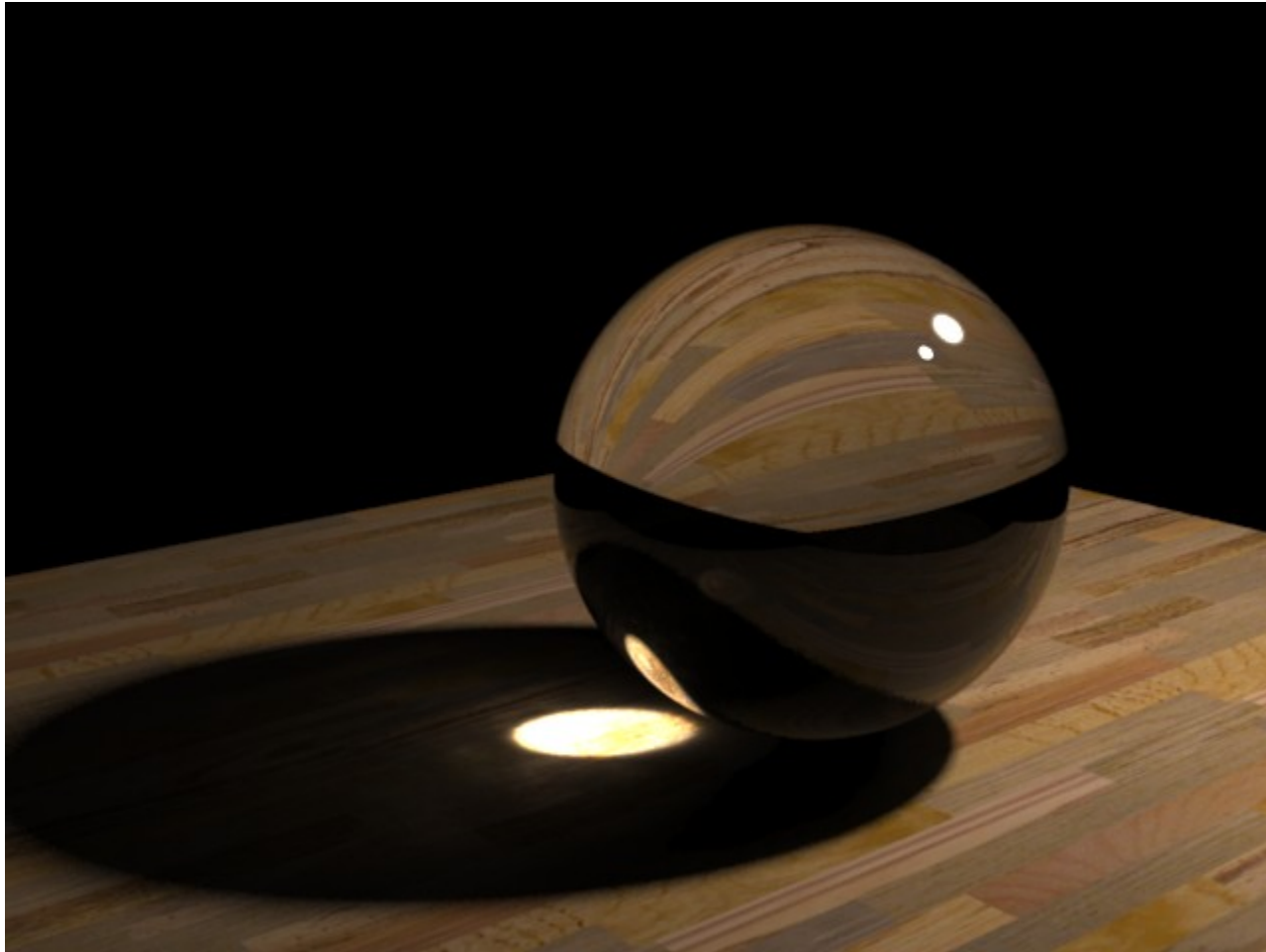
Problems with path tracing



1000 paths/pixel

[Wann Jensen]

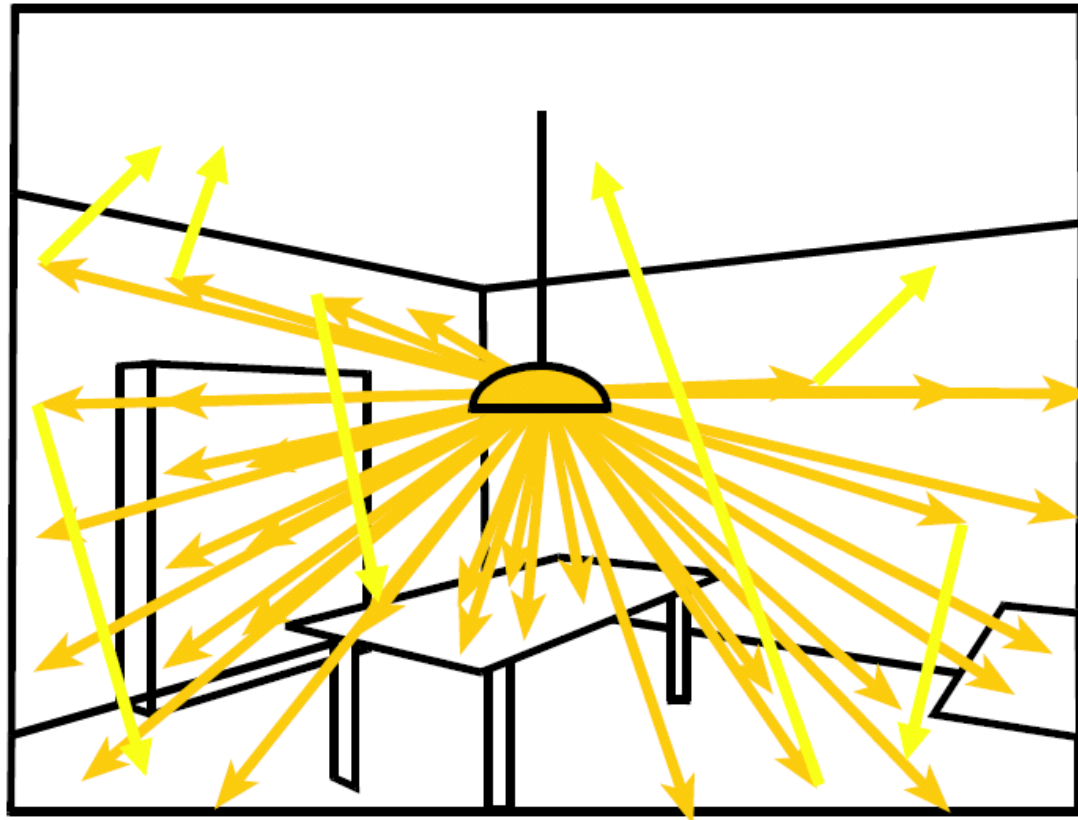
Photon mapping



[Wann Jensen]

Photon mapping

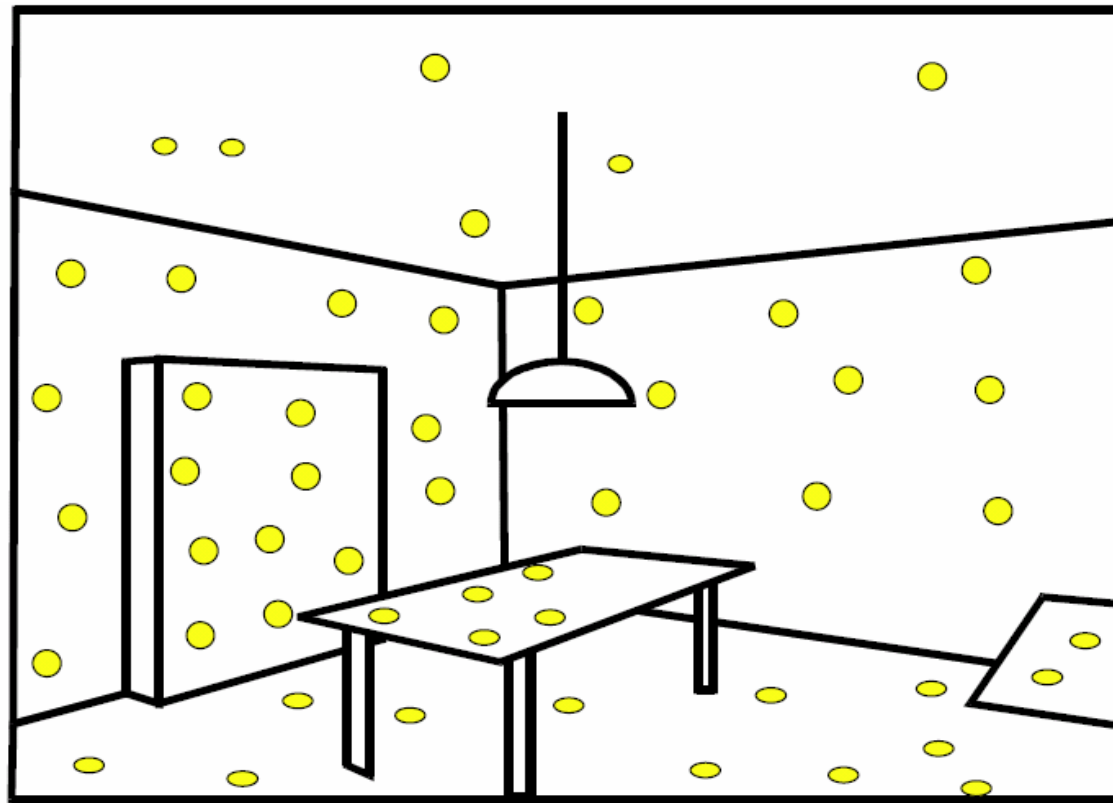
- Photon emission and transport



[Cutler, Durand]

Photon mapping

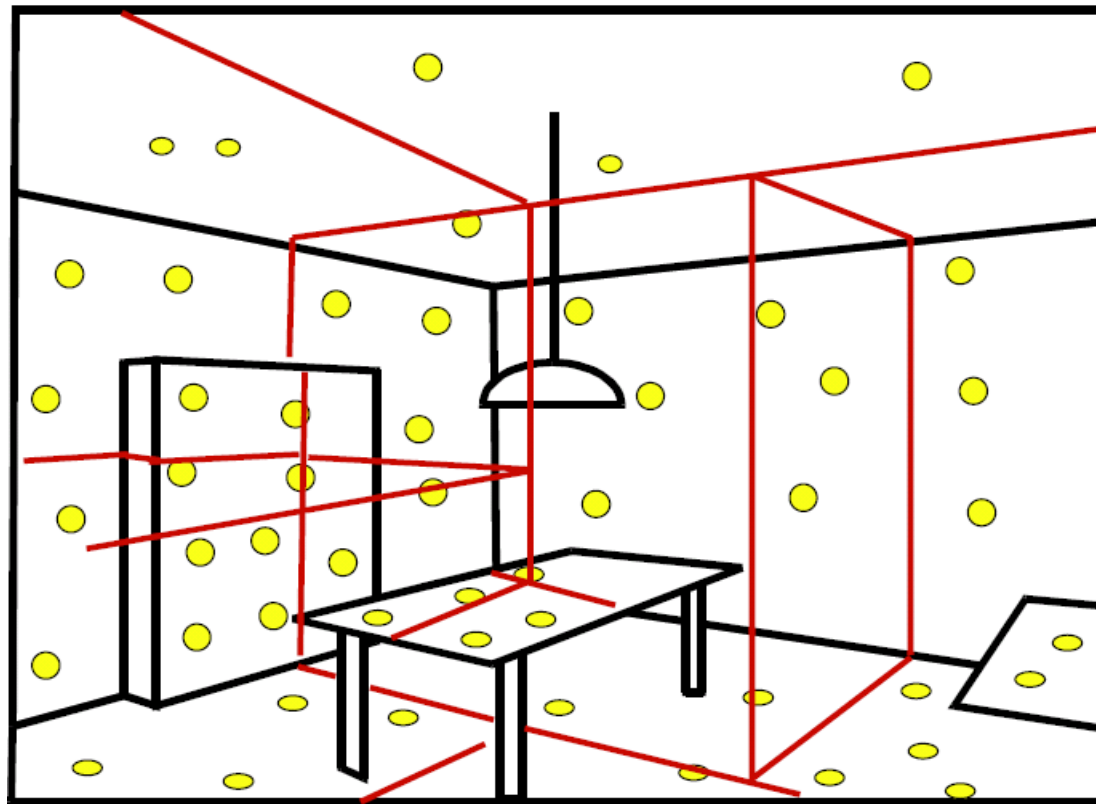
- Photon caching



[Cutler, Durand]

Photon mapping

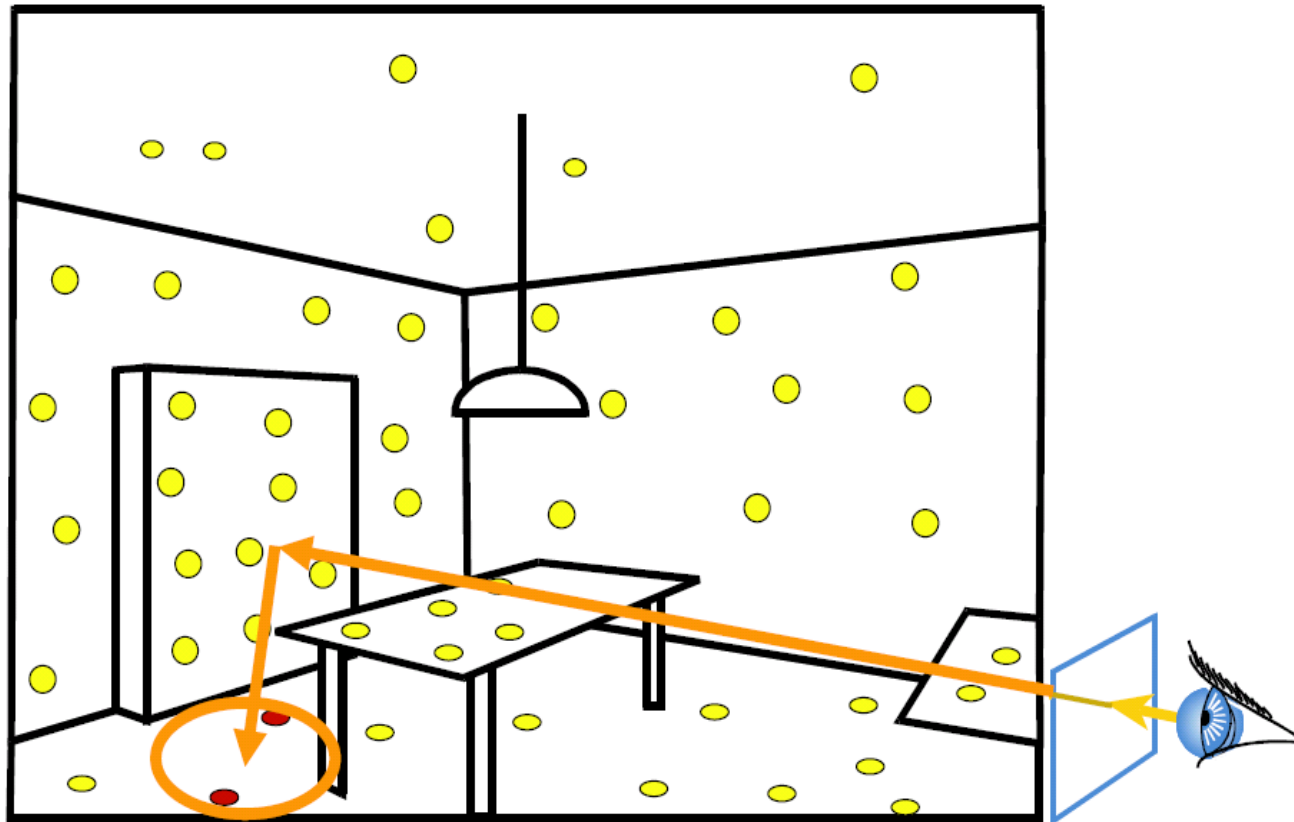
- Spatial data structure for fast access



[Cutler, Durand]

Photon mapping

- Radiance estimation



[Cutler, Durand]

What is a photon?

- A photon has a location and $\mathbf{x}$ a direction ω_i
- A photon stores
$$\alpha(\mathbf{x}, \omega_i) = \frac{L_i^*(*, \mathbf{x}, \omega_i)}{p(*, \mathbf{x}, \omega_i)}$$

such that the expected value is

$$E[\alpha(\mathbf{x}, \omega_i)] = L_i(\mathbf{x}, \omega_i)$$

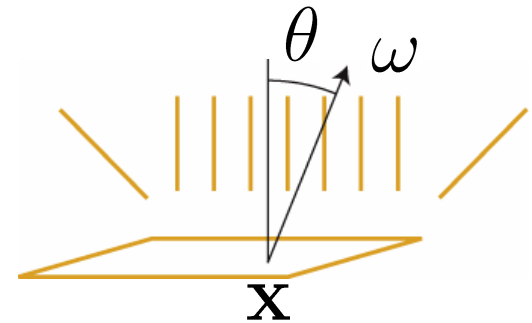
```
class photon {  
    vector3 x  
    vector2 w_i  
    vector3 a  
}
```

Photon emission

Rectangular diffuse light

- Power Φ
- Area A
- Emitted radiance $L_e(\mathbf{x}, \omega) = \Phi / (2\pi A)$
- Choose uniform PDF $p(\mathbf{x}, \omega) = 1 / (2\pi A)$
- Initialize photon

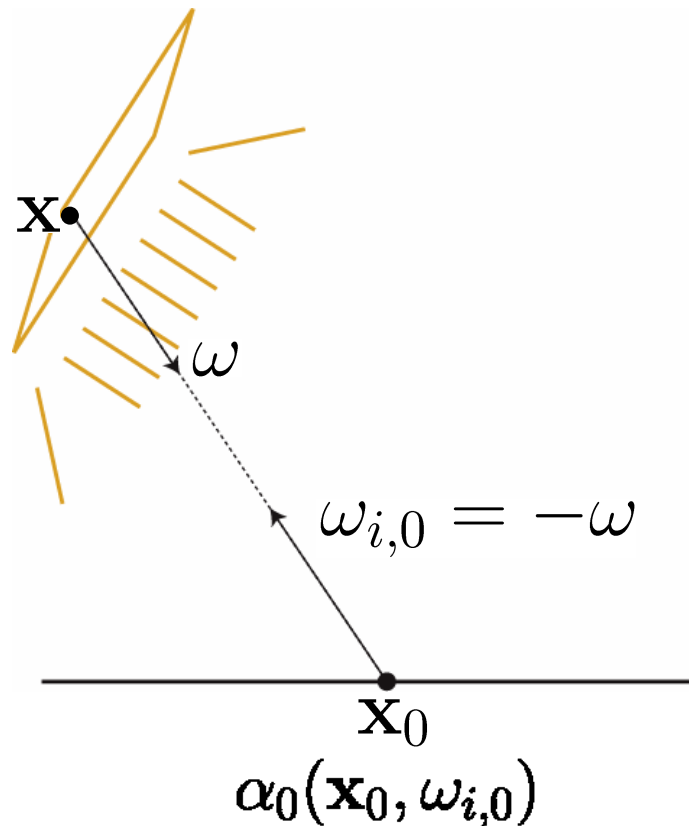
$$\alpha_0 = \frac{L_e(\mathbf{x}, \omega) \cos \theta}{2\pi A} = \Phi \cos \theta$$



Photon emission

- Store photon at first hit

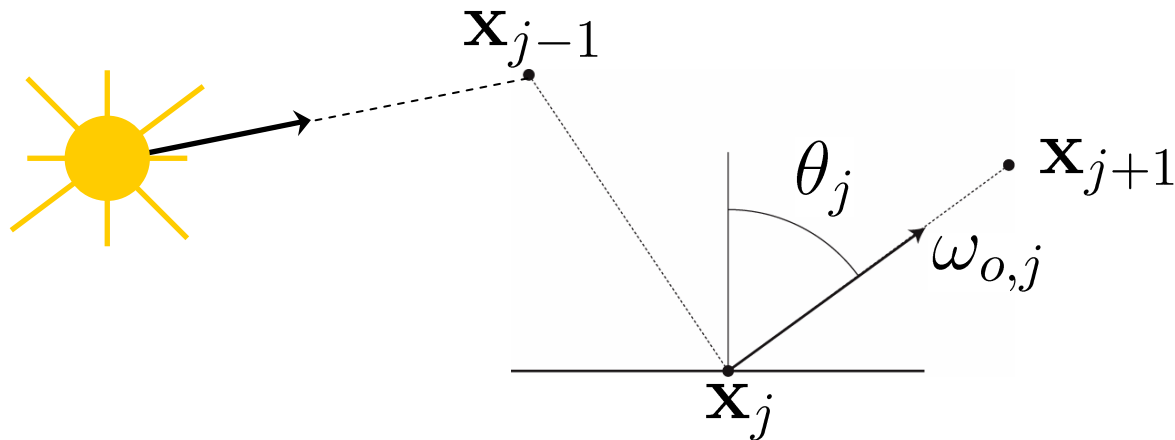
`photon_cache->add(new photon( a_0, x_0, w_i0 )`



Photon transport

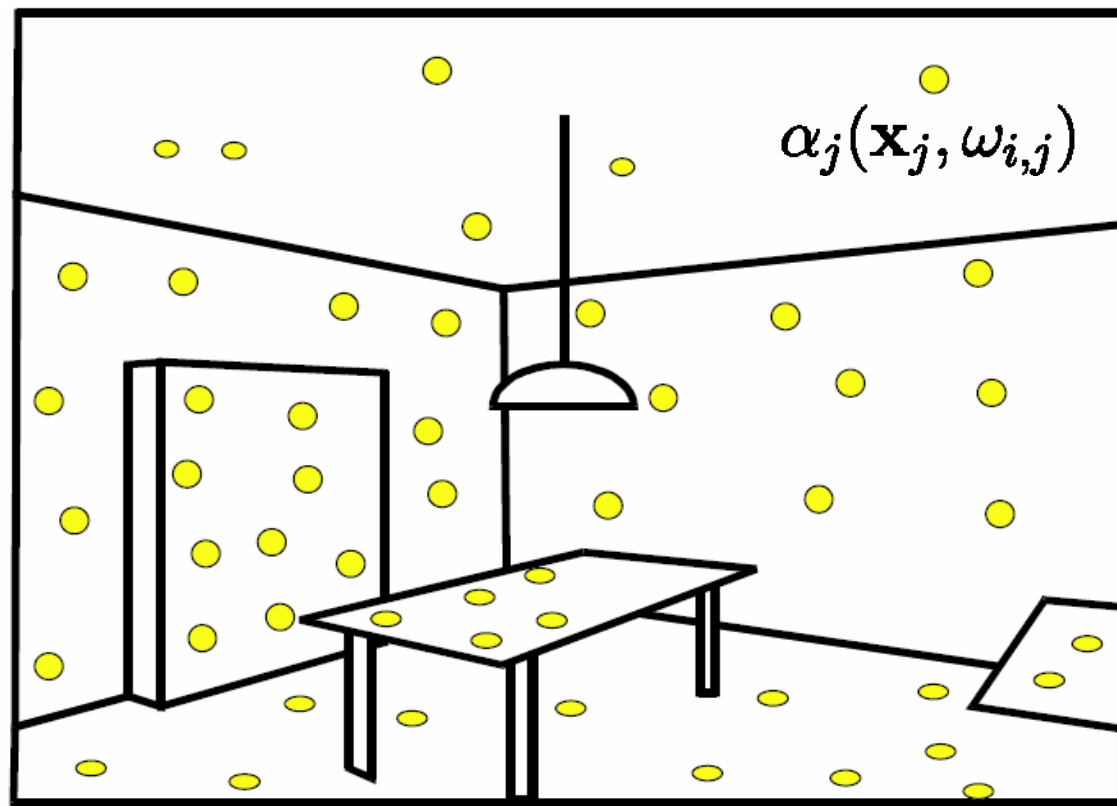
- Russian roulette with probability q_j to abort after $j - 1$ bounces
- At each surface, choose random $\omega_{o,j}$ and update photon

$$\alpha_{j+1} = \alpha_j \frac{1}{1-q_{j+1}} \frac{f(\mathbf{x}_{j-1} \rightarrow \mathbf{x}_j \rightarrow \mathbf{x}_{j+1}) \cos \theta_j}{p(\omega_{o,j})}$$



Photon mapping

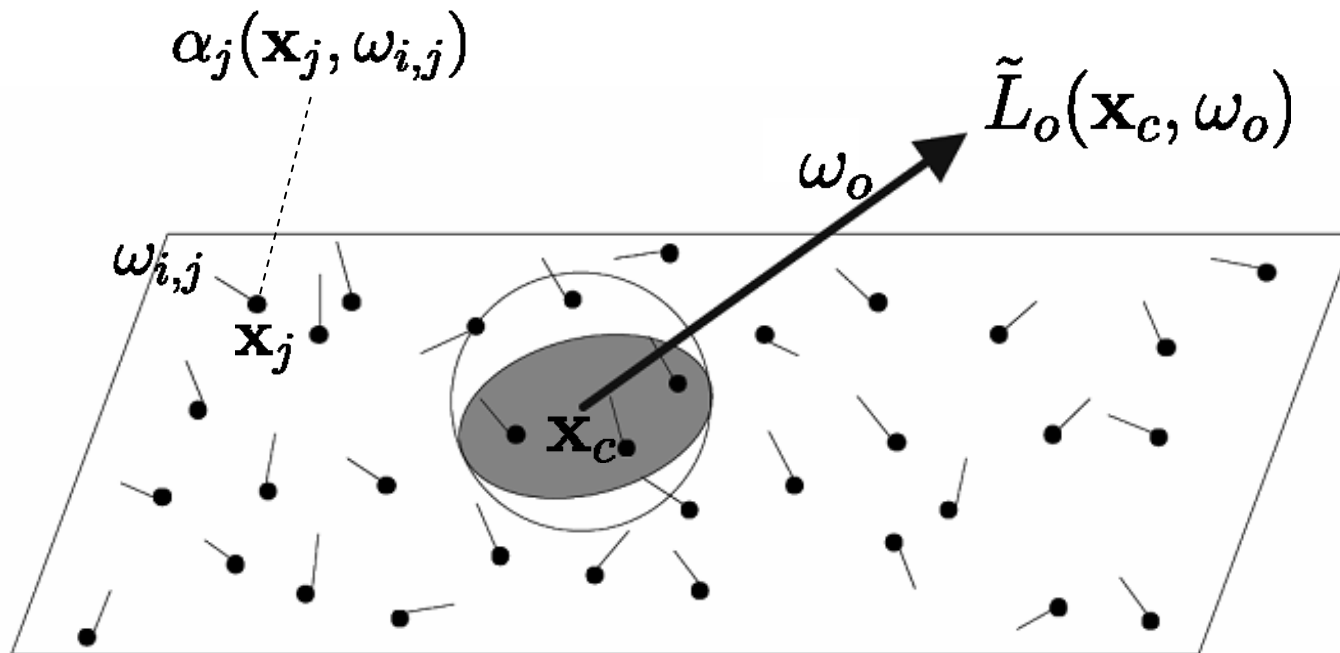
- Photon caching



[Cutler, Durand]

Taking measurements

- Use the “average” incident radiance to estimate reflected radiance



Taking measurements

- Measurement equation with “averaging kernel” W , total number of photons N

$$\begin{aligned}\tilde{L}_o(\mathbf{x}_c, \omega_o) &= \int \int W(\mathbf{x}, \omega) L_i(\mathbf{x}, \omega) d\omega d\mathbf{x} \\ &= E \left[\frac{1}{N} \sum_j W(\mathbf{x}_j, \omega_{i,j}) \alpha_i(\mathbf{x}_j, \omega_{i,j}) \right]\end{aligned}$$

- Because we made sure that

$$E[\alpha_j(\mathbf{x}_j, \omega_{i,j})] = L_i(\mathbf{x}_j, \omega_{i,j})$$

- ...introduces bias

Taking measurements

- To measure reflected radiance, we choose

$$W(\mathbf{x}_j, \omega_{i,j}) = 1/(\pi r^2) f(\mathbf{x}_c, \omega_{i,j}, \omega_o) \cos \theta_i$$

for $\|\mathbf{x} - \mathbf{x}_c\| < r$, 0 otherwise

- The estimate of the measurement is

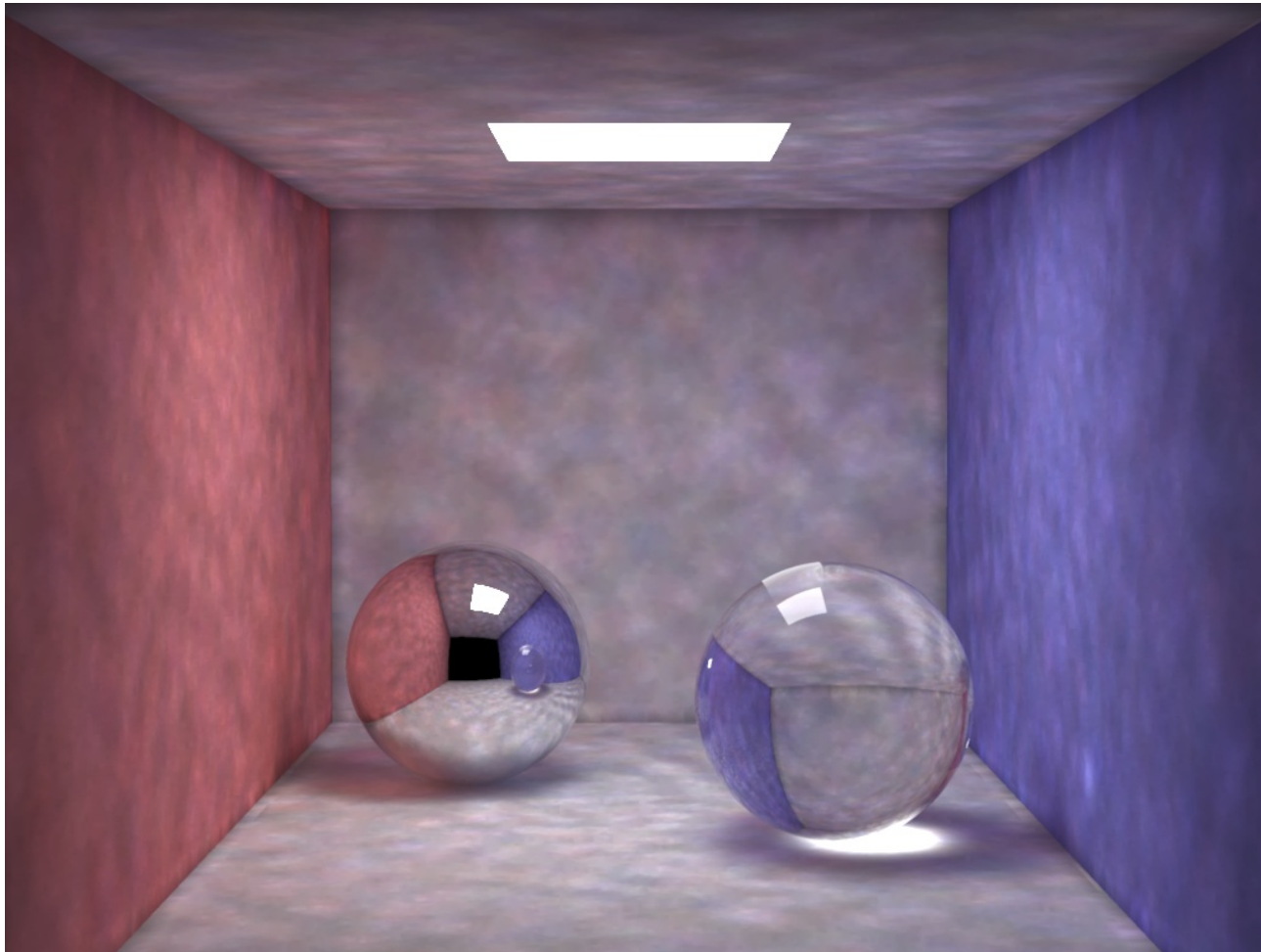
$$\tilde{L}_o(\mathbf{x}_c, \omega_o) \approx \frac{1}{N\pi r^2} \sum_{j=1}^k f(\mathbf{x}_c, \omega_{i,j}, \omega_o) \alpha_i(\mathbf{x}_j, \omega_{i,j}) \cos \theta_i$$

where $\|\mathbf{x}_j - \mathbf{x}_c\| < r$ for all $j \in 1 \dots k$

- N total paths/emitted photons

Global illumination

100000 photons, 50 photons in radiance estimate



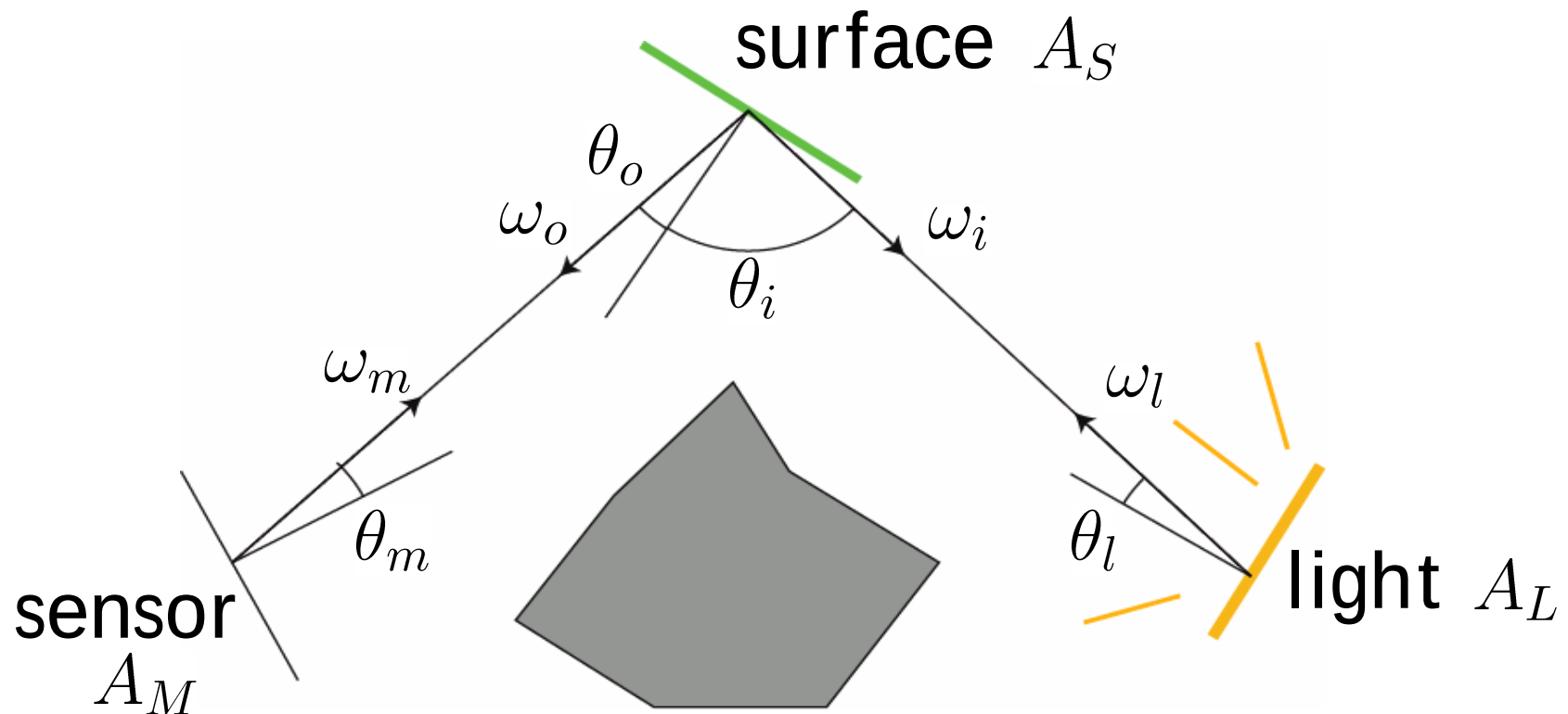
[Wann Jensen]

Today

- Explanation of cosine terms in photon transport
- Photon mapping rendering algorithms

Example

- Measure flux through sensor



Example

- Assume unit distances, constant BRDF f and constant emitted radiance L
- Power Φ

$$\Phi = \int_{A_M} \int_{\mathcal{H}^2(M)} L_i(\mathbf{x}_m, \omega_m) \cos \theta_m d\omega_m d\mathbf{x}_m$$

Example

- Assume unit distances, constant BRDF f and constant emitted radiance L
- Power Φ

$$\begin{aligned}\Phi &= \int_{A_M} \int_{\mathcal{H}^2(M)} L_i(\mathbf{x}_m, \omega_m) \cos \theta_m d\omega_m d\mathbf{x}_m \\ &= \int_{A_M} \int_{\mathcal{H}^2(M)} \left(\int_{\mathcal{H}^2(S)} f L \cos \theta_i d\omega_i \right) \cos \theta_m d\omega_m d\mathbf{x}_m\end{aligned}$$

Example

- Assume unit distances, constant BRDF f and constant emitted radiance L
- Power Φ

$$\begin{aligned}\Phi &= \int_{A_M} \int_{\mathcal{H}^2(M)} L_i(\mathbf{x}_m, \omega_m) \cos \theta_m d\omega_m d\mathbf{x}_m \\ &= \int_{A_M} \int_{\mathcal{H}^2(M)} \left(\int_{\mathcal{H}^2(S)} f L \cos \theta_i d\omega_i \right) \cos \theta_m d\omega_m d\mathbf{x}_m \\ &\approx \frac{1}{N} \sum \frac{f L \cos \theta_i \cos \theta_m}{p(\mathbf{x}_m) p(\omega_m) p(\omega_i)} \quad \text{path tracing}\end{aligned}$$

Example

$$\begin{aligned}\Phi &= \int_{A_M} \int_{\mathcal{H}^2(M)} L_i(\mathbf{x}_m, \omega_m) \cos \theta_m d\omega_m d\mathbf{x}_m \\ &= \int_{A_M} \int_{\mathcal{H}^2(M)} \left(\int_{\mathcal{H}^2(S)} f L \cos \theta_i d\omega_i \right) \cos \theta_m d\omega_m d\mathbf{x}_m\end{aligned}$$

Example

$$\begin{aligned}\Phi &= \int_{A_M} \int_{\mathcal{H}^2(M)} L_i(\mathbf{x}_m, \omega_m) \cos \theta_m d\omega_m d\mathbf{x}_m \\ &= \int_{A_M} \int_{\mathcal{H}^2(M)} \left(\int_{\mathcal{H}^2(S)} f L \cos \theta_i d\omega_i \right) \cos \theta_m d\omega_m d\mathbf{x}_m \\ &= \int_{A_M} \int_{A_S} \int_{A_L} f L \cos \theta_i \cos \theta_l d\mathbf{x}_l \cos \theta_m \cos \theta_o d\mathbf{x}_s d\mathbf{x}_m\end{aligned}$$

surface form, completely symmetric

Example

$$\begin{aligned}
 \Phi &= \int_{A_M} \int_{\mathcal{H}^2(M)} L_i(\mathbf{x}_m, \omega_m) \cos \theta_m d\omega_m d\mathbf{x}_m \\
 &= \int_{A_M} \int_{\mathcal{H}^2(M)} \left(\int_{\mathcal{H}^2(S)} f L \cos \theta_i d\omega_i \right) \cos \theta_m d\omega_m d\mathbf{x}_m \\
 &= \int_{A_M} \int_{A_S} \int_{A_L} f L \cos \theta_i \cos \theta_l d\mathbf{x}_l \cos \theta_m \cos \theta_o d\mathbf{x}_s d\mathbf{x}_m \\
 &= \int_{\mathcal{H}^2(S)} \int_{\mathcal{H}^2(L)} \int_{A_L} f L \cos \theta_l d\mathbf{x}_l \cos \theta_o d\omega_l d\omega_o
 \end{aligned}$$

Example

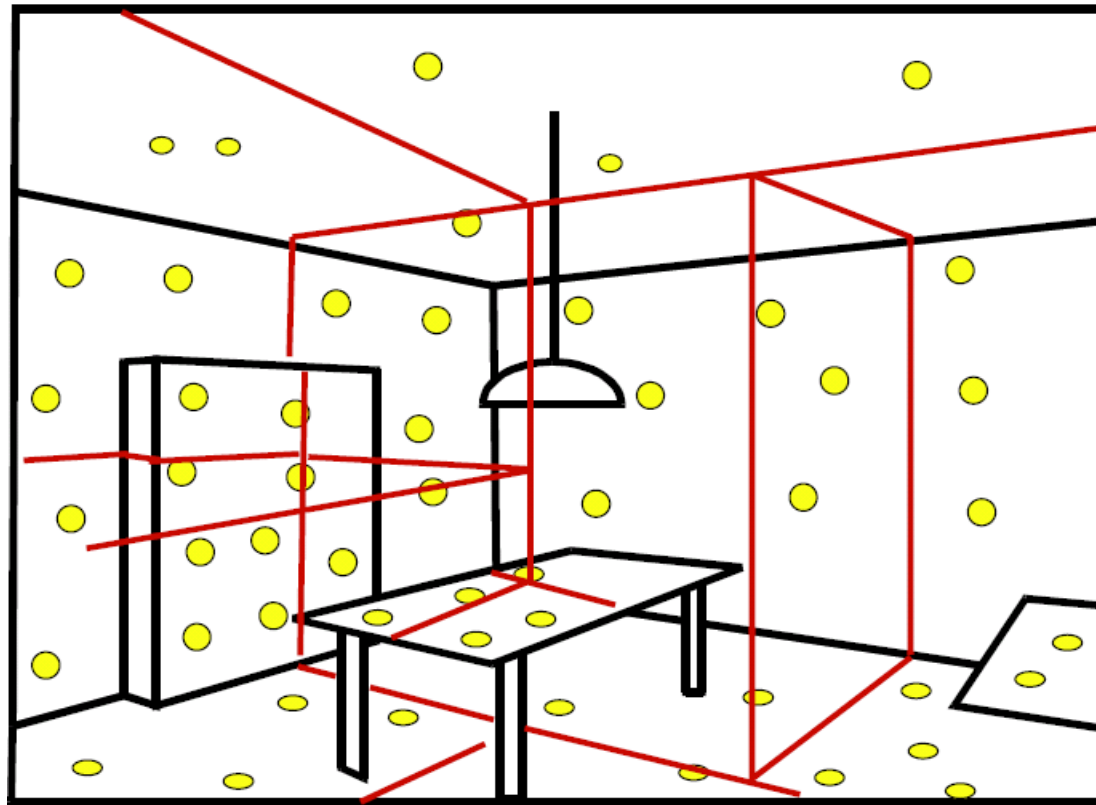
$$\begin{aligned}\Phi &= \int_{A_M} \int_{\mathcal{H}^2(M)} L_i(\mathbf{x}_m, \omega_m) \cos \theta_m d\omega_m d\mathbf{x}_m \\ &= \int_{A_M} \int_{\mathcal{H}^2(M)} \left(\int_{\mathcal{H}^2(S)} f L \cos \theta_i d\omega_i \right) \cos \theta_m d\omega_m d\mathbf{x}_m \\ &= \int_{A_M} \int_{A_S} \int_{A_L} f L \cos \theta_i \cos \theta_l d\mathbf{x}_l \cos \theta_m \cos \theta_o d\mathbf{x}_s d\mathbf{x}_m \\ &= \int_{\mathcal{H}^2(S)} \int_{\mathcal{H}^2(L)} \int_{A_L} f L \cos \theta_l d\mathbf{x}_l \cos \theta_o d\omega_l d\omega_o \\ &\approx \frac{1}{N} \sum \frac{f L \cos \theta_l \cos \theta_o}{p(\mathbf{x}_l) p(\omega_l) p(\omega_o)} \quad \text{photon mapping}\end{aligned}$$

Photon mapping algorithm

Recap

1. Emit and transport photons
2. Build data structure for fast access
3. Use stored photons to estimate radiance

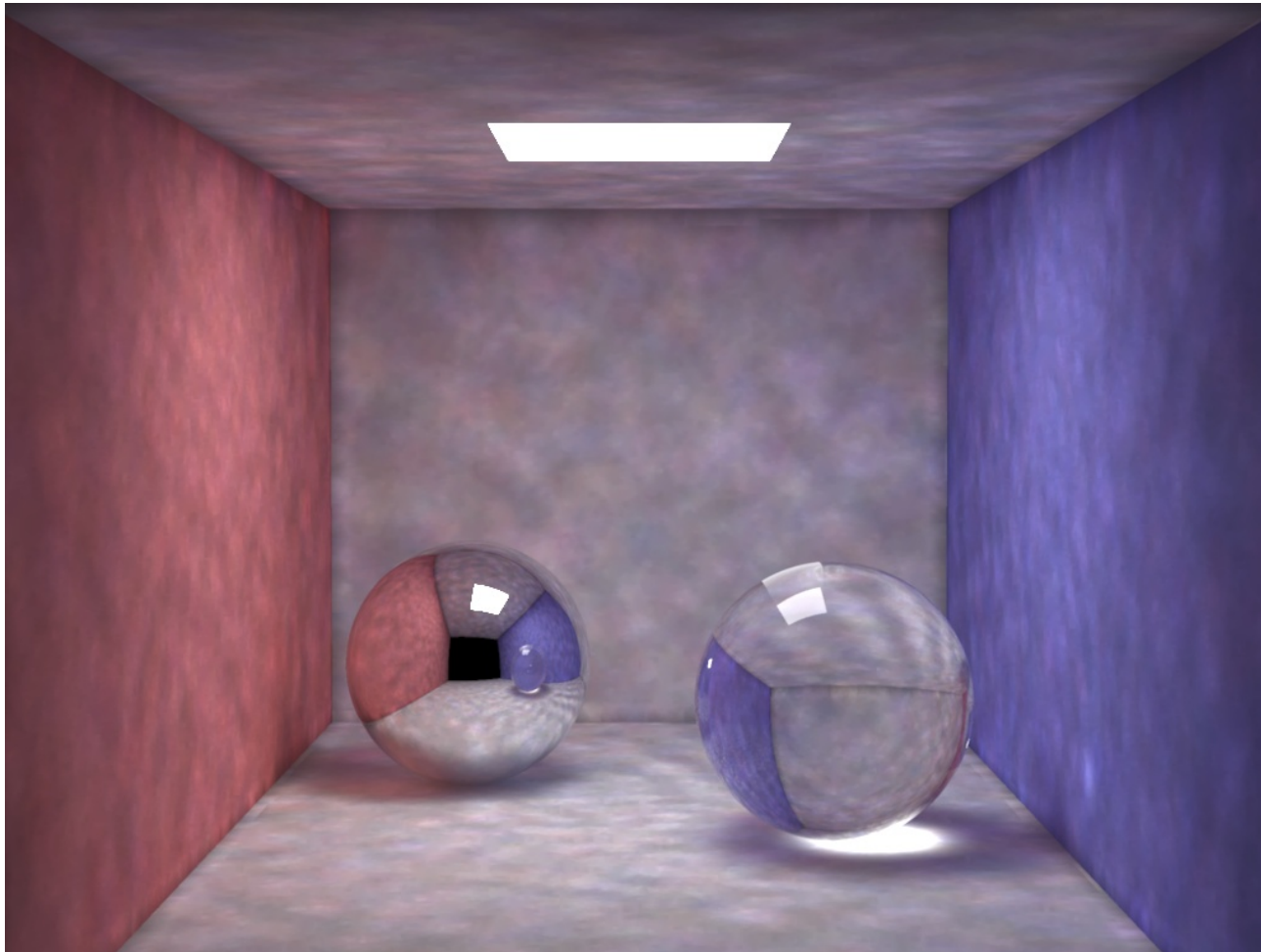
Kd-tree for fast photon access



[Cutler, Durand]

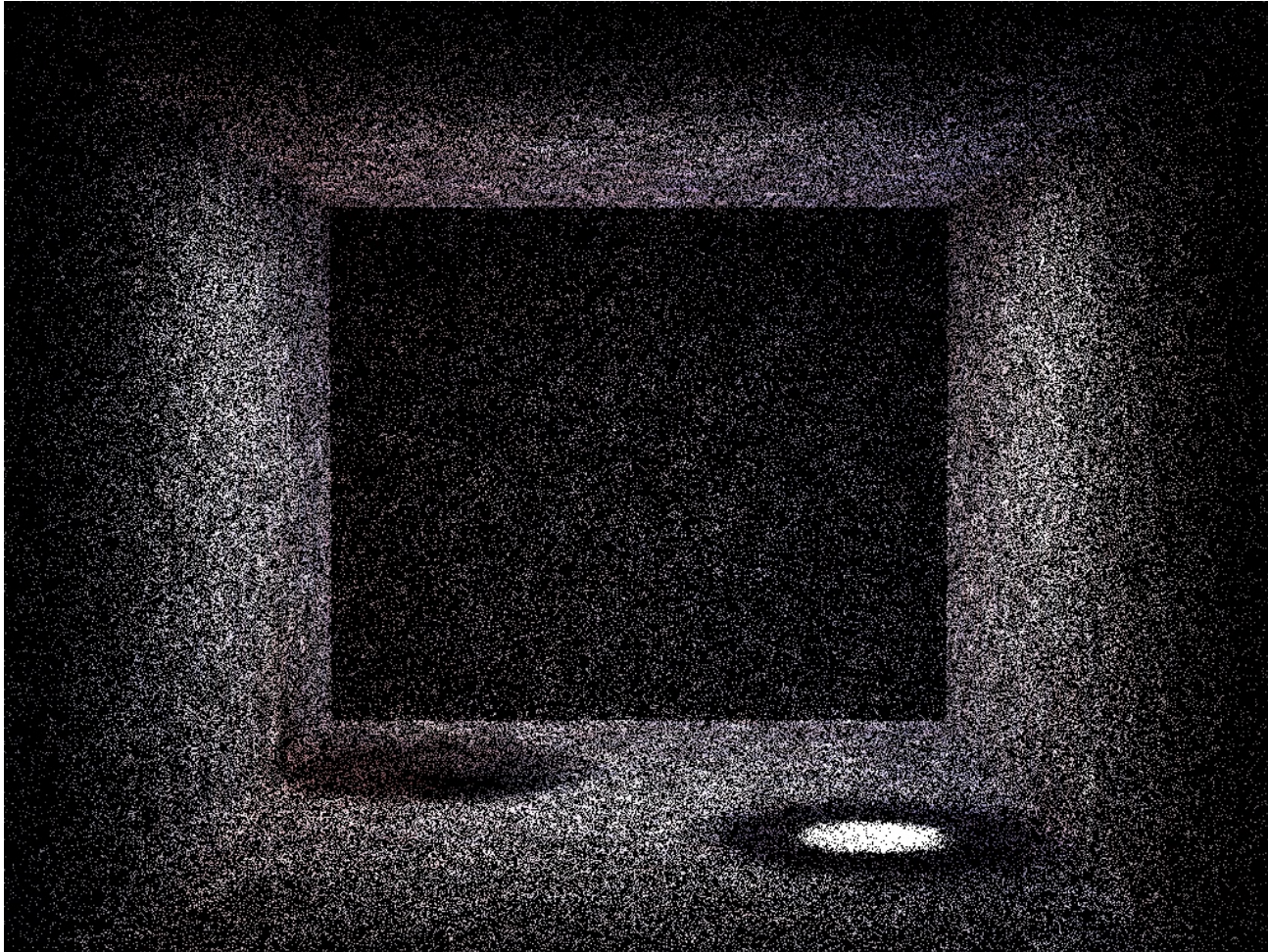
Global illumination

100000 photons, 50 photons in radiance estimate



[Wann Jensen]

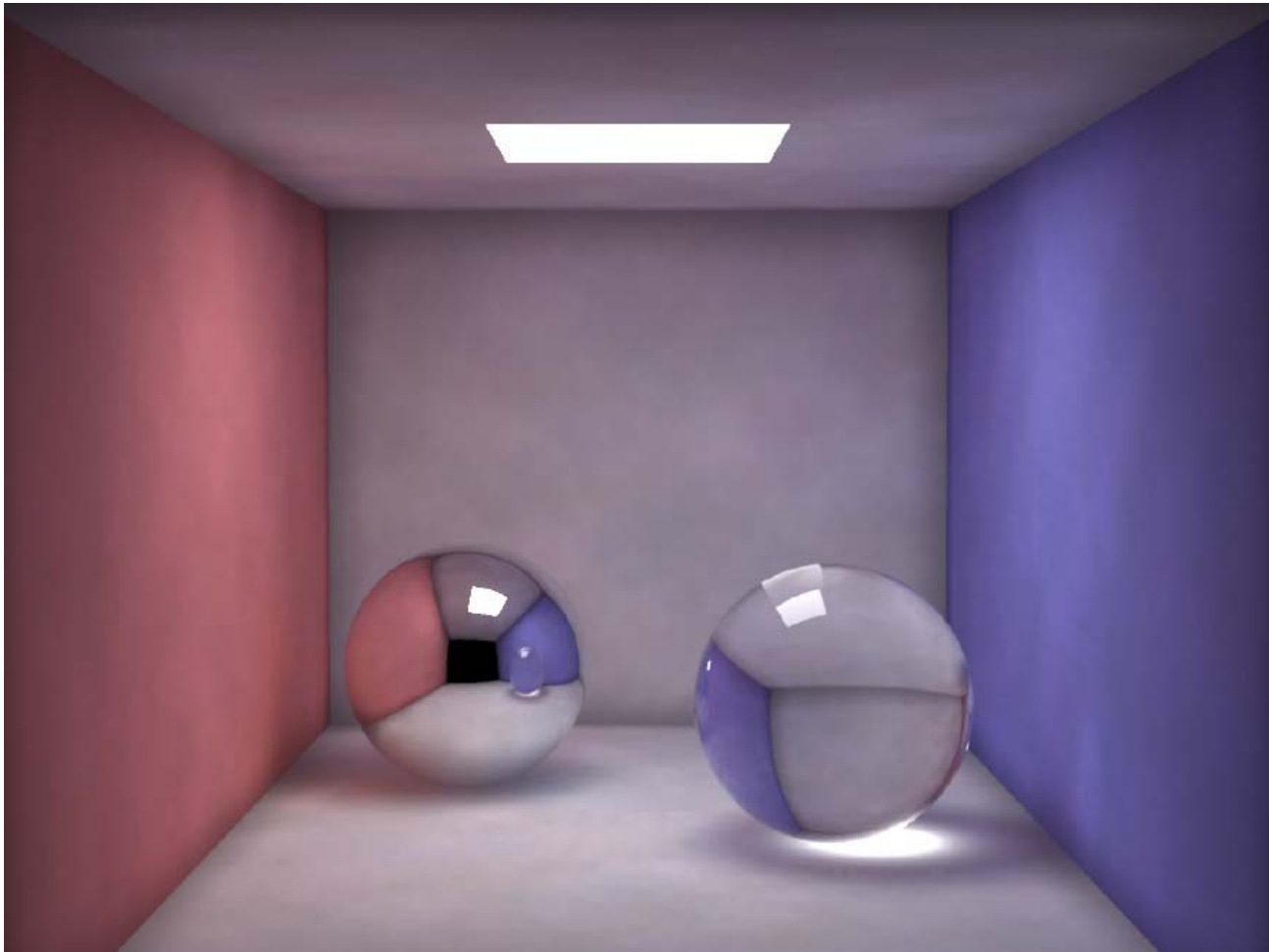
Visualization of the photons



[Wann Jensen]

Global illumination

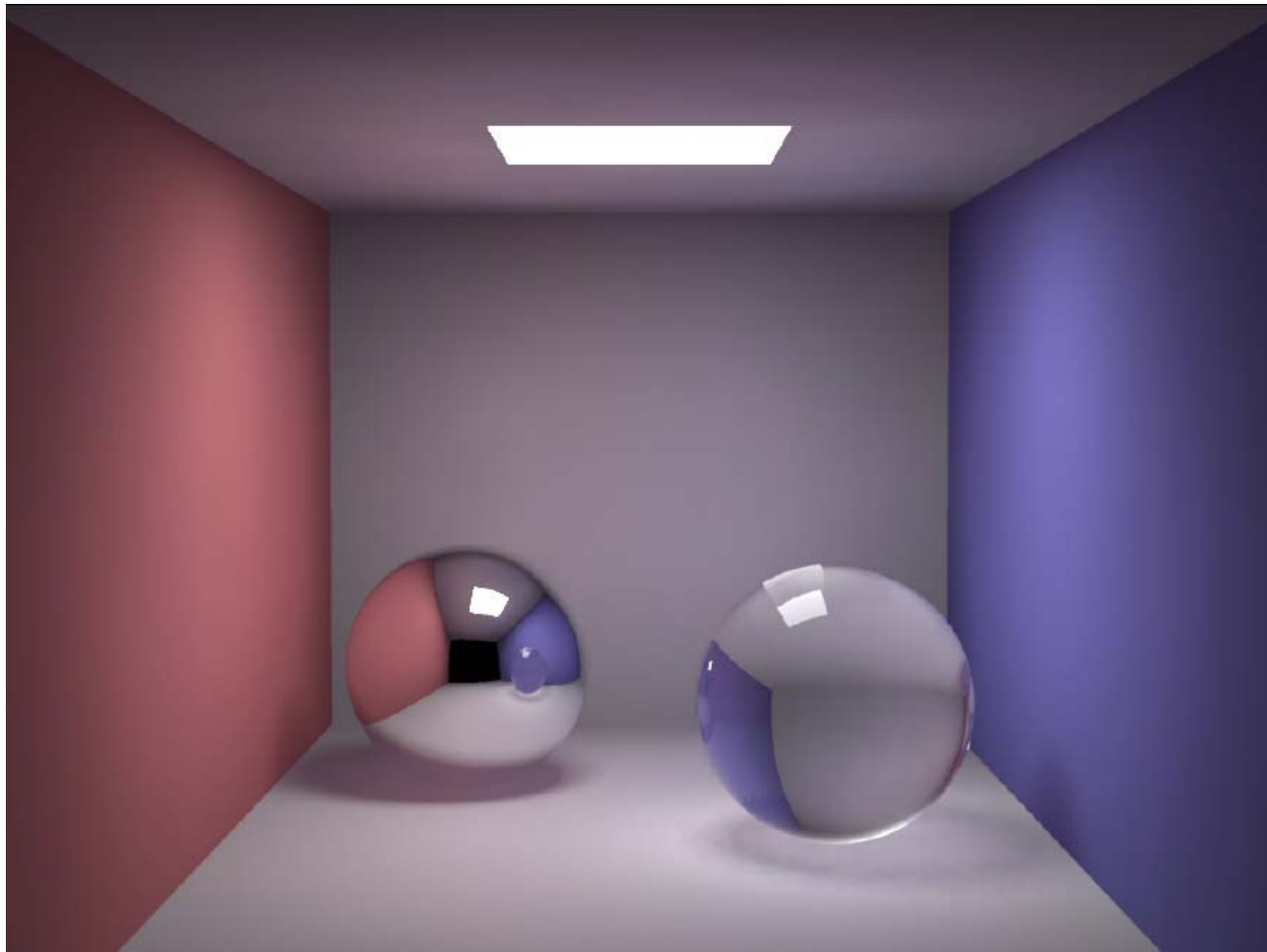
500000 photons, 500 photons in radiance estimate



[Wann Jensen]

Photons for indirect illumination

10000 photons, 500 photons in radiance estimate



[Wann Jensen]

Splitting up the reflection equation

- Reflection equation

$$L_o(\mathbf{x}, \omega_o) = \int_{\mathcal{H}^2} f(\mathbf{x}, \omega_o, \omega_i) L_i(\mathbf{x}, \omega_i) \cos \theta_i d\omega_i$$

- Split up BRDF

$$f(\mathbf{x}, \omega_o, \omega_i) = f_s(\mathbf{x}, \omega_o, \omega_i) + f_d(\mathbf{x}, \omega_o, \omega_i)$$

- Split up incoming radiance

$$L_i(\mathbf{x}, \omega_i) = L_{i,l}(\mathbf{x}, \omega_i) + L_{i,c}(\mathbf{x}, \omega_i) + L_{i,d}(\mathbf{x}, \omega_i)$$

Splitting up the reflection equation

- Reflection equation

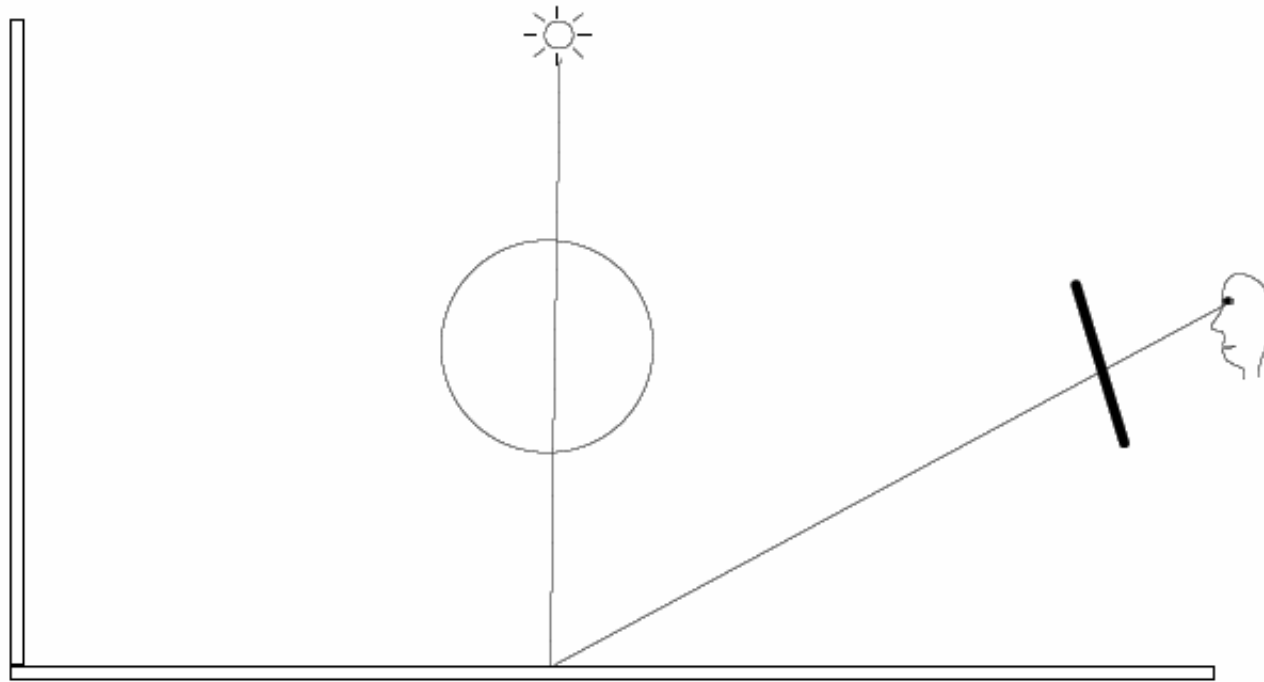
$$\begin{aligned} L_o(\mathbf{x}, \omega_o) &= \int_{\mathcal{H}^2} f(\mathbf{x}, \omega_o, \omega_i) L_i(\mathbf{x}, \omega_i) \cos \theta_i d\omega_i \\ &= \int_{\mathcal{H}^2} f(\mathbf{x}, \omega_o, \omega_i) L_{i,l}(\mathbf{x}, \omega_i) \cos \theta_i d\omega_i \\ &\quad + \int_{\mathcal{H}^2} f_s(\mathbf{x}, \omega_o, \omega_i) (L_{i,c}(\mathbf{x}, \omega_i) + L_{i,d}(\mathbf{x}, \omega_i)) \cos \theta_i d\omega_i \\ &\quad + \int_{\mathcal{H}^2} f_d(\mathbf{x}, \omega_o, \omega_i) L_{i,c}(\mathbf{x}, \omega_i) \cos \theta_i d\omega_i \\ &\quad + \int_{\mathcal{H}^2} f_d(\mathbf{x}, \omega_o, \omega_i) L_{i,d}(\mathbf{x}, \omega_i) \cos \theta_i d\omega_i \end{aligned}$$

- Evaluate each part separately, each with different algorithm

Photon maps

- Three photon maps, one for each illumination term
 - Direct
 - Indirect diffuse
 - Caustic (indirect specular)
- Finite state machine to keep track of photon type

Direct illumination



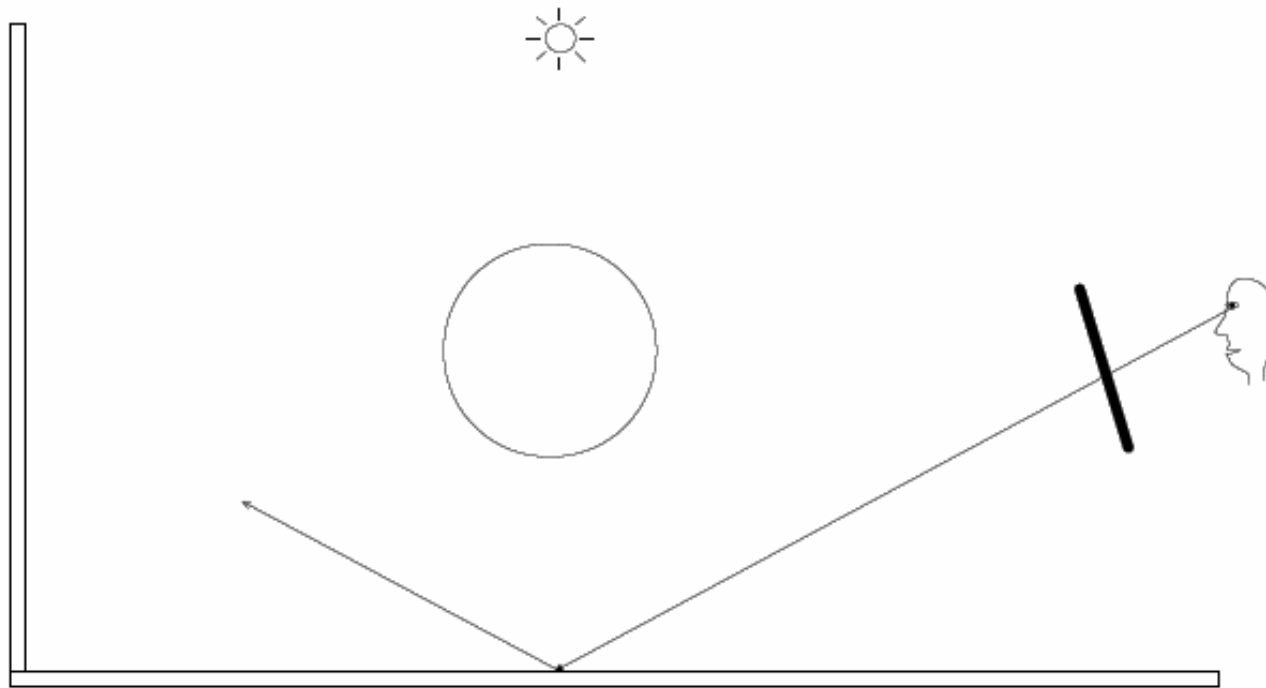
[Wann Jensen]

Direct illumination

$$\int_{\mathcal{H}^2} f(\mathbf{x}, \omega_o, \omega_i) L_{i,l}(\mathbf{x}, \omega_i) \cos \theta_i d\omega_i$$

- Path tracing
- Sample light sources
- Multiple importance sampling

Specular reflection



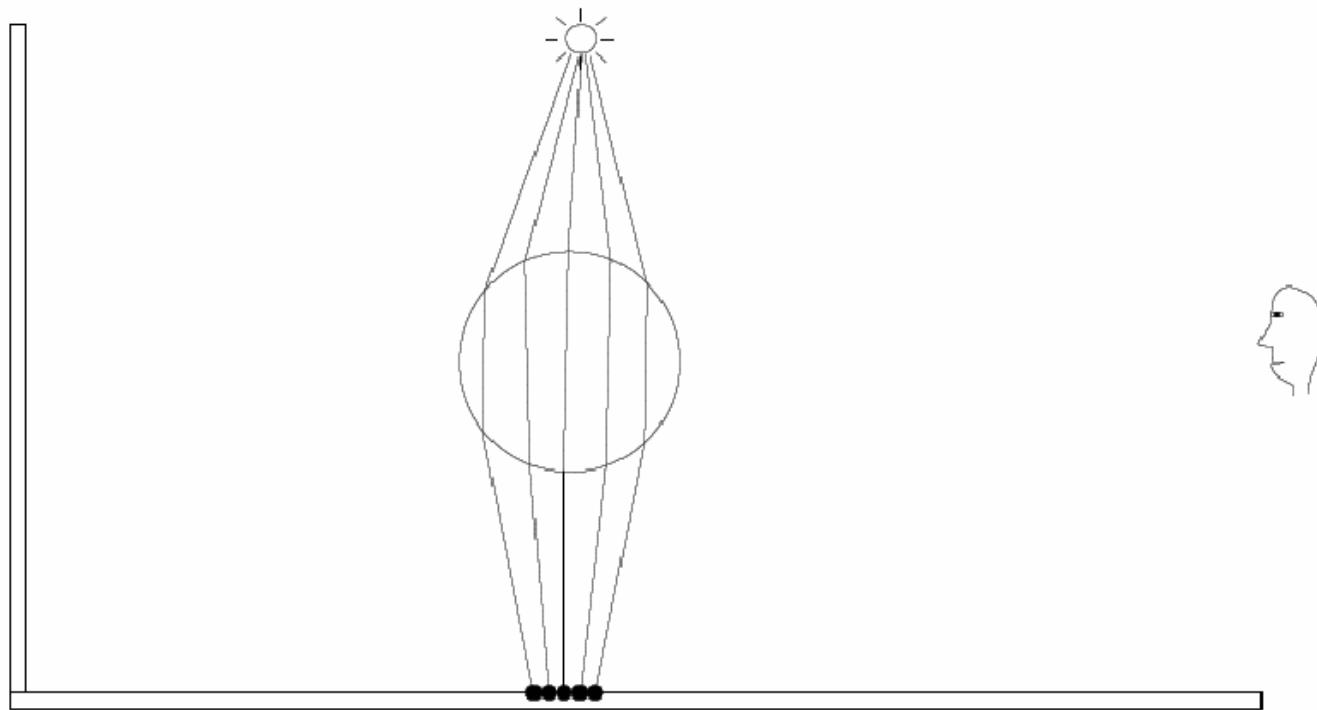
[Wann Jensen]

Specular reflection

$$\int_{\mathcal{H}^2} f_s(\mathbf{x}, \omega_o, \omega_i) (L_{i,c}(\mathbf{x}, \omega_i) + L_{i,d}(\mathbf{x}, \omega_i)) \cos \theta_i d\omega_i$$

- Path tracing
- No need to store photons on purely specular surfaces

Caustics



[Wann Jensen]

Caustics

$$\int_{\mathcal{H}^2} f_d(\mathbf{x}, \omega_o, \omega_i) L_{i,c}(\mathbf{x}, \omega_i) \cos \theta_i d\omega_i$$

- Indirect illumination due to purely specular paths (LS+D paths)
- Radiance estimation using caustic photon map

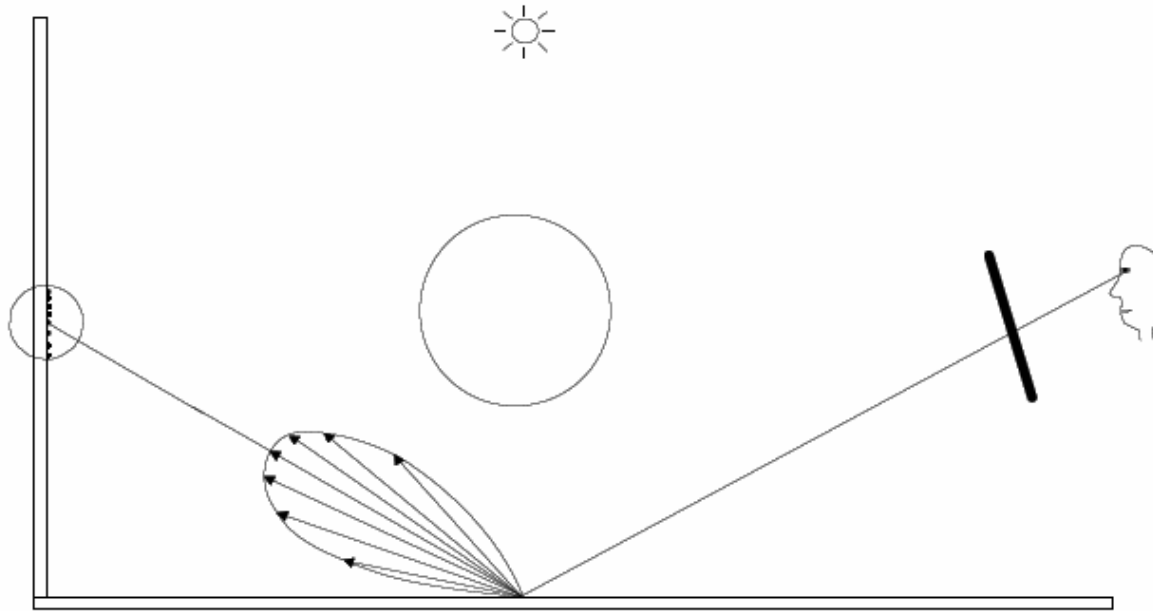
Diffuse indirect illumination

$$\int_{\mathcal{H}^2} f_d(\mathbf{x}, \omega_o, \omega_i) L_{i,d}(\mathbf{x}, \omega_i) \cos \theta_i d\omega_i$$

- Indirect illumination that has bounced at least once off a diffuse surface
- Accurate: path tracing
- Less accurate, often good enough: final gathering using all three photon maps
- Even less accurate: radiance estimation using indirect photons

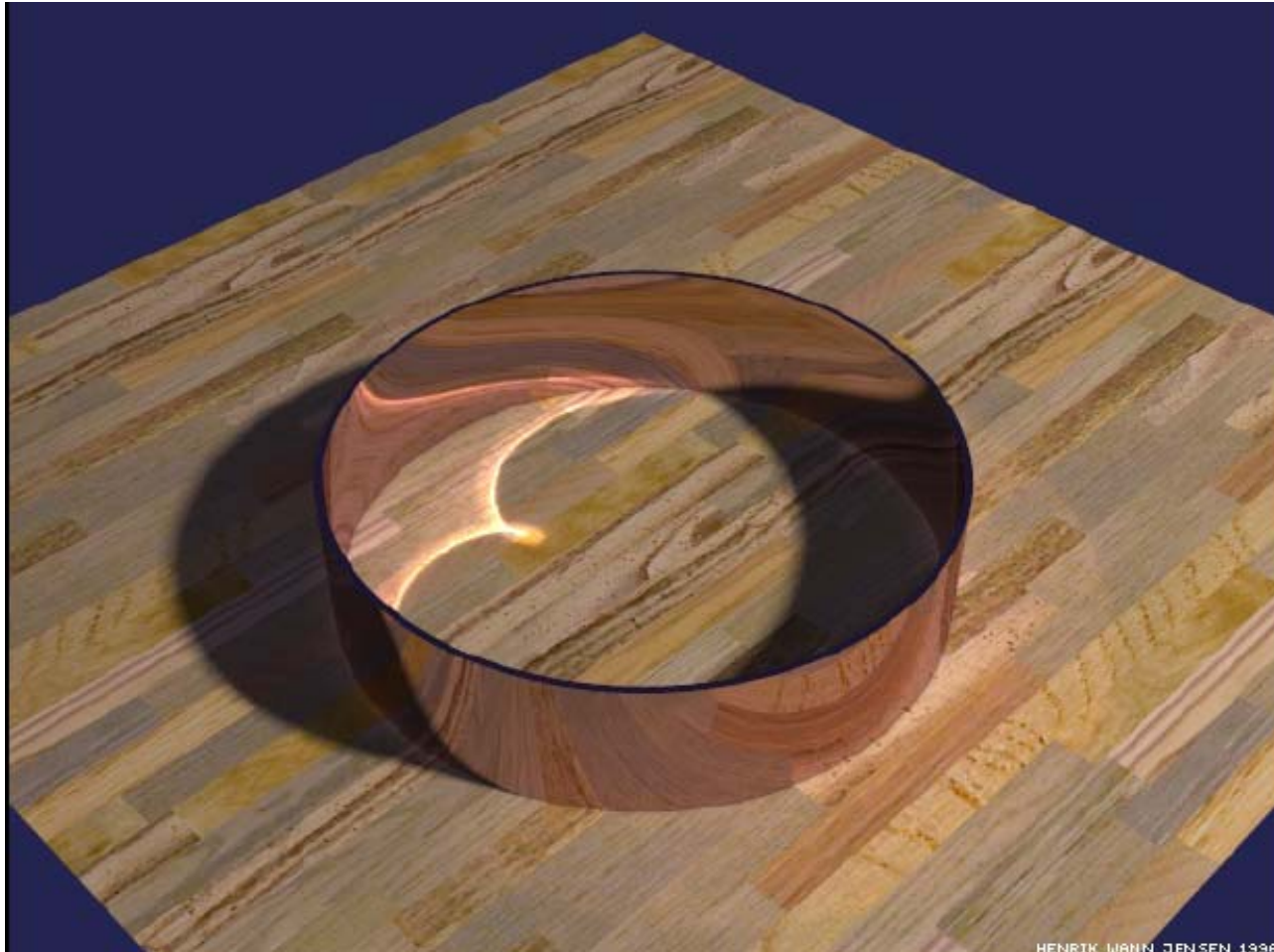
Diffuse indirect illumination

- Final gathering



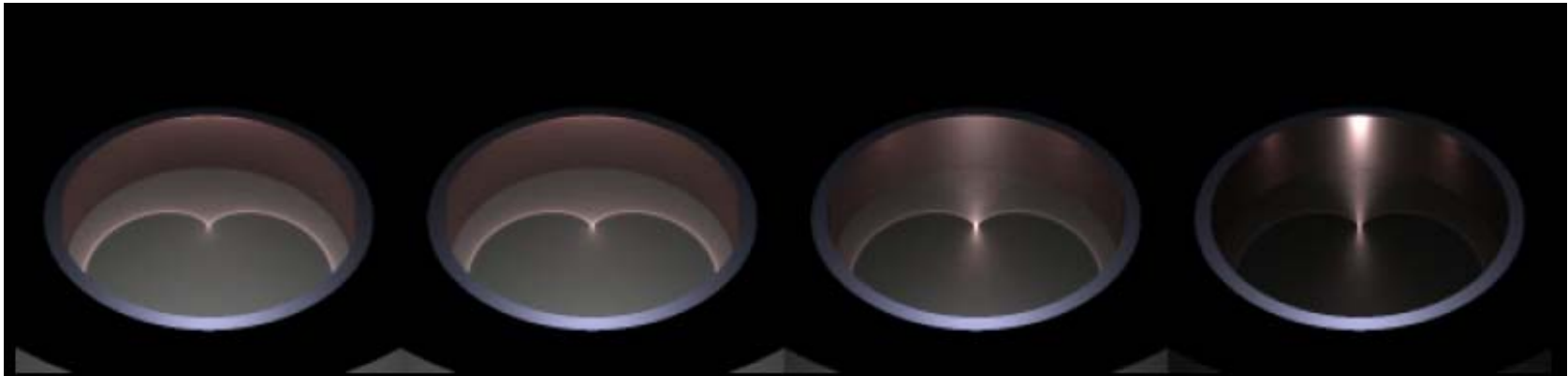
[Wann Jensen]

Reflections inside a ring



50000 photons, 50 photons for radiance estimation

Caustics on glossy surfaces



[Wann Jensen]

340000 photons, ~100 photons for radiance estimation

HDR environment illumination



[Wann Jensen]

Lightprobe from www.debevec.org

Next time

- Radiosity